

dlbs Mini-Challenge

February 21, 2026

1 General Remarks on the Assignment

Choose a problem statement and use case in image and/or signal processing for this mini-challenge. The task should be solvable with deep learning (object detection, image segmentation, or signal prediction). The solution approach follows established methodological guidelines [Kar19, Lon23].

1.1 Collaboration

Submissions are typically in teams of two. Exceptions (e.g., individual work) require approval. Collaboration across teams is allowed, but each team must submit unique work.

1.2 Milestones and Deadlines

1. **M1:** MC proposal
2. **M2:** Learning exchange (progress up to overfitting stage)
3. **M3:** Individual check-in
4. **M4:** Final submission

Deadlines are listed in the Spaces calendar.

1.3 Written Submissions and Deliverables

1. **MC proposal (at M1):** Problem statement, use case, research questions, data situation, and methodological approach (1–2 pages; template: Sect. 3).
2. **Final Report (at M4):** Results synthesis in German or English (6–8 pages; maximum word count binding; template: Sect. 4).
3. **Notebooks and Code (at M4):** HTML export of relevant notebooks and a Git repository link to the source code. No training/test data to be submitted. The solution process must be traceable; selected data must be visualized or made playable (details: Sect. 4–6).
4. **Presentation Material (at M4):** Slides, if applicable.
5. **Self-Assessment (at M4):** Self-assessment of the achieved grade, including a reflective justification.

1.4 Learning Exchange (M2)

Each team delivers a 10-min demo/presentation followed by 5-min Q&A and provides structured feedback to another team:

- Strengths: What worked particularly well?
- Areas for improvement: What could have been clearer or more convincing?
- Key takeaway: What did I take away from this presentation?

The demo/presentation includes a recap of the problem statement and use case, dlbs-specific exploratory data analysis, and a baseline overview of achieved performance with respect to overfitting.

2 Assessment

General rules:

- If a model does not learn the target, no points will be awarded.
- Late submissions are rejected.
- Content exceeding the specified word count per section will not be read or graded.

See grading rubric for assessment (xlsx file).

3 Proposal and Final Solution

For **M1**, the MC proposal should be concise and focused, emphasizing synthesis, abstraction, and decision-making rather than detailed exposition. The domain context and the rationale for the chosen data and methodological approach must be clearly stated (1–2 pages).

For **M4**, revise the proposal to reflect the final data and methods.

3.1 Problem Statement / Use Case (Report)

*TASK: Describe the problem statement and the use case you are addressing, each in one sentence. Formulate 1-2 explicit and quantitative hypotheses, research questions, or objectives. This section defines the **what** of your mini-challenge. (150 words)*

Guidelines:

- *The problem statement should describe the technical task to be solved.*
- *The use case should explain the application context and motivation.*
- *Hypotheses or objectives should be measurable and testable.*
- *Use precise terminology and avoid implementation details.*

3.2 Data (Report)

TASK: Choose a dataset that allows you to solve your problem statement or use case. You may use images (2D, 3D), videos, sensor signals, point clouds, audio data, or similar. The dataset can be publicly available, your own, or from a business context. Since this MC addresses a supervised problem, ground truth annotations or labels must be available or generated with minimal effort. Small datasets are sufficient for the MC; the implications are discussed in text. (150 words)

Guidelines:

- *State the dataset size and structure (e.g. number of samples, classes).*
- *Describe data type and resolution (e.g. image size, sampling rate).*
- *List relevant features and available annotations (including format).*
- *Provide the data source or reference (link, repository, or origin). Data itself is not shared.*
- *Discuss limitations, especially if the dataset is small or imbalanced.*

3.3 Methods (Report)

TASK: Describe and discuss the (choice of) methods you use(d) to address your problem statement/use case. This includes your overall solution approach with the chosen models or architectures, your baseline, the evaluation strategy, and key parameters. Explain why you selected a particular approach, method, metric, or parameter for a specific objective. Include a sketch or figure illustrating your workflow. (300 words)

Guidelines:

- *Focus on goal-oriented reasoning: why a certain approach for which objective.*
- *Use present tense for general statements, past tense for implementation details.*
- *Discuss baselines, models, evaluation metrics, and key parameters.*
- *Keep the section concise and well structured.*

4 Results

Results are documented via implementation notebooks and the written report: notebooks ensure traceability through step-by-step experimentation, parameter studies, and diagnostics, while the report synthesizes concepts, methods, and final outcomes. Implementations should be provided in a Jupyter notebook, a Weights & Biases (WandB) report, or a comparable format. Notebooks must be readable and well structured; excessive code output should be hidden or made scrollable where appropriate.

4.1 Data Inspection (Notebook)

*TASK: Perform a **dlbs-specific** exploratory data analysis following these steps [Kar19] and have each person discuss one step in 2-3 sentences:*

- *Inspect the data visually*
- *Develop a qualitative understanding of the data*
- *Analyze the data quantitatively (using code)*

Guidelines: Follow for the data inspection guidelines from [Lon23]:

- *Do take the time to understand your data*
- *Don't look at all your data*
- *(Do make sure you have enough data)*
- *Do talk to domain experts*
- *Do survey the literature*
- *Do think about how your model will be deployed*
- *Don't allow test data to leak into the training process*

4.2 Setup a Baseline (Notebook)

TASK: Define and evaluate a simple baseline model, selecting and justifying appropriate evaluation metrics and parameters. The baseline may be trained, generated, or calculated, and should be analyzed and visualized (e.g., loss curves, predictions). Each team member discusses one related aspect from [Kar19] in 2-3 sentences.

Guidelines: Follow baseline modeling guidelines from [Lon23]:

- *Do use an appropriate test set*
- *Do use a validation set*
- *Do save some data to evaluate your final model instance*
- *Don't use accuracy with imbalanced data sets*
- *Do evaluate a model multiple times*
- *Don't do data augmentation before splitting your data*
- *Don't allow test data to leak into the training process*

4.3 Overfit (Notebook)

TASK: Select a model architecture appropriate to the research question and deliberately overfit the model following [Kar19]. Focus on the training loss, gradually introduce additional model or training complexities, and document their effects. Each team member implements or executes a different overfitting-related aspect from [Kar19] and discusses the results in 2–3 sentences. If a model fails to learn, this issue must be resolved before proceeding further. Optional bonus: Compare and evaluate an additional deep learning architecture or alternative model.

Guidelines: When overfitting models, follow the guidelines from [Lon23]:

- *Don't use inappropriate models*
- *Do try out a range of different models*
- *Do evaluate a model multiple times*
- *Don't do data augmentation before splitting your data*

4.4 Regularize (Notebook)

TASK: Improve generalization performance by increasing validation accuracy, potentially at the expense of training accuracy. Each team member conducts 1–2 regularization experiments following [Kar19] and discusses the observed effects in 1–2 sentences per regularization aspect. Compare the resulting models using appropriate evaluation procedures.

Guidelines: When comparing models, follow the evaluation and reporting guidelines from [Lon23]:

- *Do evaluate a model multiple times*
- *Don't assume a bigger number means a better model*
- *Do use statistical tests when comparing models*
- *Do correct for multiple comparisons*
- *Don't always believe results from community benchmarks*
- *Do be transparent*
- *Do report performance in multiple ways*
- *Do be careful when reporting statistical significance*
- *Do look at your models*

4.5 Tune (Notebook)

TASK: Continue with the selected best model and perform hyperparameter tuning using one of the following approaches:

- *Random search*
- *Grid search*
- *Hyperparameter optimization*

Analyze and document the tuning process and results.

Guidelines: When tuning models, follow the guidelines from [Lon23]:

- *Do evaluate a model multiple times*
- *Do optimize your model's hyperparameters*
- *Do be careful where you optimize hyperparameters and select features*

4.6 Most Relevant Results (Report)

TASK: Select your most important results and describe them. Use figures where appropriate (e.g. particularly good vs. poor model predictions). (450 words)

Guidelines:

- *Focus on results that directly address the research question(s)/objectives/use case.*
- *Use figures selectively and ensure they are clearly explained in the text.*
- *Report results objectively; interpretation and implications belong in the discussion.*

5 Discussion (Report)

TASK: Discuss your results explicitly in relation to your research question(s)/objective(s). Evaluate how well your model addresses the problem statement and use case, including the meaning of errors and performance scores. Compare baseline results with optimized models and analyze which cases work well and which do not. Conclude whether the research question can be answered with the chosen models and data. (450 words)

Guidelines: For the discussion, follow the guidelines from [Lon23]:

- *Do not generalize beyond the available data.*
- *Do not uncritically trust results from community benchmarks.*
- *Be transparent about assumptions and limitations.*

In practice, this means:

- *Interpret results in the context of the use case, not only in terms of scores or methods.*
- *Compare baseline and optimized models critically.*
- *Discuss advantages, disadvantages, opportunities, and risks of your approach.*
- *Explain the relevance and limitations of benchmark results.*
- *Clearly state uncertainties, failure cases, and constraints.*

6 Reflection (Report)

TASK: Provide a final technical reflection and a self-assessment. Technical reflection: What conclusions can be drawn for the specific problem statement and use case? What worked well in this mini-challenge, and what would you do differently next time? What would you change in the assignment? Self-assessment: self-assess your grade according to the rubric and briefly justify each assessment category. (150 words)

Guidelines: Focus on technical insights and lessons learned rather than project management.

7 Code

TASK: Provide access to a tidy Git repository containing your code, a README file, and one main file that exported to HTML. Remove obsolete files or move them to an archive. The submission must not contain commented-out code sections.

Guidelines:

- *Ensure the repository is well structured and easy to navigate.*
- *Grant the instructor access to the repository:*
 - *GitHub: `susuter`*
 - *GitLab: `susanne.suter@fhnw.ch`*

- *The main file should be available as an HTML export (e.g. Jupyter Notebook, Quarto, or equivalent) and serve as the primary entry point for the project.*
- *Write code that is self-explanatory and readable.*
- *Use comments sparingly and only to explain why a decision was made, why a calculation was performed in a specific way, or which source was used. My recommendation is: as many comments as necessary, but as few as possible.*
- *Remove commented-out code; such code is considered obsolete and not tested (except for `pip install` statements).*

References

- [Kar19] Andrej Karpathy. A recipe for training neural networks, 2019.
- [Lon23] Michael A. Lones. How to avoid machine learning pitfalls: a guide for academic researchers, 2023.